

SWAPNIL SHAH

Software Development Engineer | Java / Spring Boot / React / Svelte
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SUMMARY

Software engineer with **6+ years** building and owning production systems end to end — backend architecture, data layers, workflow orchestration, and the React, Svelte 5, TypeScript, and JavaScript frontends on top. Shipped AI document processing handling **230K+ invoices/month** at **75% lower cost (\$85K/year saved)** and **99% first-pass accuracy**. Sole owner of a B2B payments module processing **\$75M+ in annual volume**. Led a Hibernate-to-jOOQ migration across 100+ PostgreSQL tables and moved 25+ jobs onto **Temporal** workflows across **AWS EKS and GCP GKE**, serving **5,000+ hospitality locations**.

TECHNICAL SKILLS

Languages	Java (8+), Python, TypeScript, JavaScript, SQL / PL-pgSQL, C++
Backend & Frameworks	Spring Boot, Spring MVC / Security / Data / AOP, jOOQ, Hibernate / JPA, Temporal, REST, OpenAPI, GraphQL, Apache Kafka, Node.js / Express, JUnit, Mockito
Frontend	Svelte 5, React.js (Hooks, Redux), Angular, Ember.js, TypeScript codegen, HTML5, CSS3, ESLint
AWS Cloud	EKS, Aurora PostgreSQL, ElastiCache, S3, DynamoDB, ECR, EC2, Lambda, VPC / ALB / ACM / Route53, Secrets Manager
GCP & Other Cloud	GKE, Google Cloud Platform, Azure, Firebolt
DevOps & IaC	OpenTofu / Terraform, Docker, Kubernetes, GitHub Actions, Jenkins, SOPS, OIDC auth
Data & Storage	PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, Redis, Qdrant (vector), DynamoDB
AI / ML	Vision-language models (Kimi, Gemini), BGE-VL embeddings, PaddleOCR, vector retrieval, RAG pipelines, pandas, NumPy, scikit-learn
Observability & Tools	New Relic, JMX, HikariCP tuning, Git, GitLab, Bitbucket, IntelliJ IDEA, VS Code, Jira

PROFESSIONAL EXPERIENCE

Craftable	Sep 2024 – Present
Senior Software Engineer	
- Led modernization of a legacy Java monolith into a modular Spring Boot service backend, migrating 50+ REST endpoints across Analytics, Logbook, Books, and Reports domains and cutting p95 API latency ~40%. - Migrated the persistence layer from Hibernate to jOOQ across 20+ backend modules and 100+ PostgreSQL tables, eliminating N+1 query patterns and reducing report query times up to 60% on multi-million-row datasets while enabling compile-time SQL type safety. - Authored 80+ Flyway migrations, including a zero-downtime store-cleanup program executed in marshall-safe SQL slices against live production traffic with no maintenance window. - Re-architected 25+ long-running third-party operations (POS FTP sweeps, Gmail ingestion, Finexio payment reconciliation, Firebolt uploads, DSR sync) from blocking synchronous execution onto asynchronous, non-blocking Temporal workflows across AWS EKS and GCP GKE, so vendor API timeouts no longer cascade into failed jobs; added automatic retries, deterministic replay, and orphan detection; cut daily reconciliation MTTR from hours to under 5 minutes. - Co-architected the Invoice AI OCR platform serving 5,000+ hospitality locations (230K+ invoices/month, 800K+ lifetime): migrated a costly Gemini-only pipeline to a hybrid vision-language system (BGE-VL visual embeddings, Qdrant historic-invoice retrieval, PaddleOCR structure extraction) spanning AWS (Java/Temporal) and GCP (Python/GKE), then to self-hosted Kimi open-source VLMs for extraction; cut per-invoice cost 75% (\$85K/year saved) and lifted first-pass accuracy from 90% to 99%. - Owned and operated a multi-cloud production stack: AWS (EKS across a multi-account Organization, VPC/ALB/ACM/Route53, Aurora PostgreSQL, ElastiCache, ECR, S3, DynamoDB, Secrets Manager) and GCP GKE for AI services; authored OpenTofu modules and GitHub Actions workflows with OIDC cloud auth, environment-gated production deploys, and SOPS-encrypted secrets, and tuned Hikari connection pools with JMX metrics exported to New Relic. - Sole owner of Craftable Pay (Finexio B2B payments), architecting and shipping the module end-to-end from vendor onboarding and check-run submission through refund handling and accounting reconciliation; processes \$75M+ in annual payment volume. - Co-architected the Ember.js to Svelte 5 frontend rewrite serving 5,000+ hospitality locations, defining component-library conventions, the OpenAPI-driven TypeScript codegen pipeline, page-level styling rules enforced via custom ESLint, and a v2 error-contract standard adopted across the frontend team. - Shipped 10+ enterprise POS integrations end-to-end (UKG Ready Labor, Shift4/SkyTab to QuickBooks DSR, CBS NorthStar/NSOE, Toast, Lightspeed Series K, SpotOn, Agilysys, Oracle Symphony) on a shared parser-registry framework, onboarding thousands of hotel and restaurant locations.	

- Delivered and hardened **5+ enterprise accounting integrations** (Microsoft Dynamics 365 tax and approval, NetSuite SuiteTalk SOAP bill-payment sync, Sage Intacct, Acumatica, QuickBooks Online), unblocking top-tier customer accounts and **eliminating 20,000+ manual reconciliation entries per month**.
- Owned the **Operator AI** insights feature end to end: **Temporal** batch and schedule workflows aggregating 60-day rolling POS, invoice, and waste data into a Python scoring service, jOOQ persistence across 7 insight tables, and the **Svelte 5** frontend — scored insight cards, evidence charts, and detail drawers surfacing ML-ranked sales, labor, and cost opportunities with a **human-in-the-loop feedback loop** capturing structured operator ratings for model improvement.

New Jersey Institute of Technology | Newark, NJ

Sep 2023 – May 2024

Teaching Assistant

- Teaching Assistant for **CS332 (Principles of Operating Systems)**, supporting students on process scheduling, memory management, and concurrency.
- Held office hours and delivered one-on-one guidance to help students debug assignments and work through complex systems concepts.
- Graded assignments, quizzes, and exams for CS332 and **CS114 (Introduction to Computer Science II)**, maintaining accuracy, consistency, and adherence to grading rubrics.

Fluvina Technologies | India

May 2019 – Aug 2022

Software Developer

- **Directed the government data ingestion platform end to end**, designing, building, and maintaining **Python ETL pipelines** against government-authorized APIs (XML, REST, SOAP) using requests, pandas, and lxml, normalizing heterogeneous agency schemas into a unified PostgreSQL model and **cutting large-dataset processing time 80%**.
- Sustained and hardened that ingestion layer in **Python** across three years of upstream API changes, adding schema-drift detection, deduplication, type coercion, and idempotent retry/backoff handling, converting a recurring manual reprocessing burden into unattended scheduled runs.
- Built **Python** data-preparation and feature-engineering workflows (pandas, NumPy, scikit-learn) to identify key predictive variables for patient risk and health-scoring models, delivering analysis-ready datasets to the ML effort.
- **Engineered and operated backend infrastructure end to end**, from Spring Boot and Express services on PostgreSQL and AWS EC2 through Redis caching, Jenkins CI/CD, and production troubleshooting, **reducing downtime 25%** and cutting request latency via Java performance tuning and memory profiling.
- **Drove Mobihealth from requirements through release and post-launch support**, an application suite serving healthcare professionals, patients, receptionists, and pediatric users across four distinct role-based experiences.
- **Architected and maintained the React.js frontend** for Mobihealth, establishing a reusable component library with Hooks-based state management and Redux for cross-module data flow, standardizing form, table, and scheduling primitives shared across all four portals.
- Built the **React** Patient Relation Management dashboard and a configurable personalized home screen, surfacing vitals history, risk scores, and follow-up state while rendering dynamically from user preferences — **lifted user satisfaction 25%**.
- Implemented vital-tracking features in **Zero Sugar**, driving a **20% improvement in health outcomes** for diabetic patients.
- **Spearheaded software delivery end to end for MAHI**, an IoT pill-dispensing device built on **core Python and PyQt5**, architecting synchronized online and offline modes with local queuing and reconciliation on reconnect, **reducing device downtime 40%**.
- Designed microservices on Spring Boot and Spring Cloud and enforced testing discipline (Spring Boot Test, Mockito, JUnit) that sustained code quality across three years of releases.

Technicra IT Solutions | India

May 2018 – Apr 2019

Software Developer

- Built Spring Boot microservices and REST APIs with **Apache Kafka** as the inter-service messaging backbone, applying Spring Core (IoC), AOP, and JDBC, with Hibernate for persistence.
- Engineered multi-threaded server-side components using Java concurrency primitives (thread pools, ExecutorService, synchronized shared state) to handle concurrent requests without blocking the request path, improving responsiveness under peak load.
- Developed serverless microservices as **AWS Lambda** functions in Node.js, and deployed server-side modules and client-side interfaces on **AWS EC2**.
- Consolidated multi-page flows into a single-page application using **Angular and TypeScript**, alongside HTML, CSS, jQuery, and React.js work.
- Developed and maintained legacy **JSP, Servlet, and Struts** modules on **IBM WebSphere**, including JDBC data-access components retrieving customer account and transaction records from **Oracle**.
- Wrote unit and integration tests with **JUnit and Spock**, and performed debugging and performance profiling across release cycles.

EDUCATION

Master of Science, Computer Science

New Jersey Institute of Technology, Newark, NJ

Bachelor of Engineering, Computer Engineering

Dharmsinh Desai University, Nadiad, Gujarat